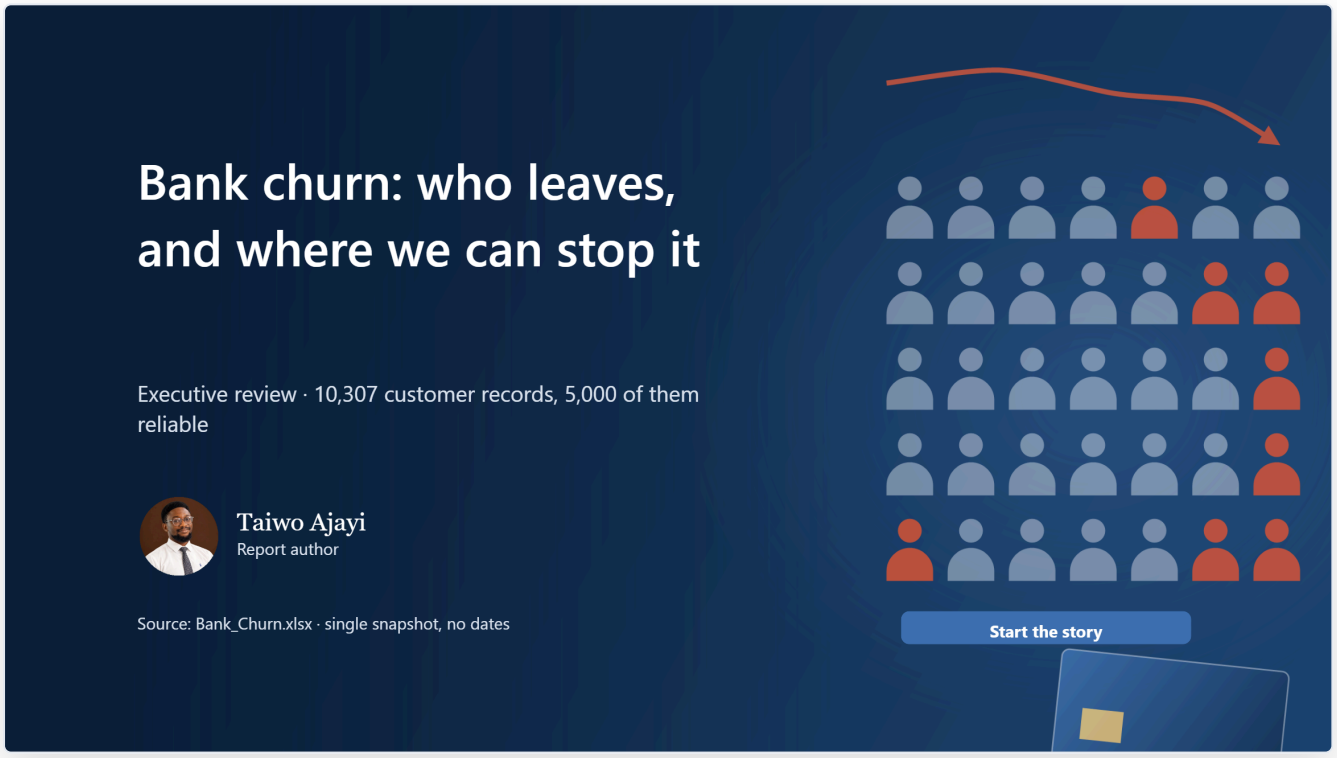
PORTFOLIO CASE STUDY · POWER BI

Bank churn: who leaves, and where we can stop it

A retail bank in three markets is losing customers. I found who leaves, why, and which two actions would keep the most of them.



Taiwo Ajayi
Data analyst · Power BI developer



DOMAIN
Retail banking

DATA
10,307 customer records (5,000 reliable), 1 snapshot

MODEL
Star schema: 1 fact, 8 dimensions, 42 measures

REPORT
15 pages: 11 story, 3 dashboards, 1 drill-through

Power BI · DAX · Power Query · Data modelling · Data quality · Storytelling

THE PROJECT

The challenge

The bank believed churn was random, that more products and higher balances made customers stickier, and that country and gender mattered. Management needed to know which beliefs held, who actually leaves, and where retention money would work.

The data

One Excel workbook with a customer table and six lookup sheets. Profiling showed the file splits at row 5,000: the later 5,307 rows churn at a flat 51% in every group and contradict their own band keys in 98% of cases. I based every finding on the 5,000 reliable rows and kept the rest, flagged, for transparency.

My approach

1. Profiled the workbook and found the reliable / suspect split at row 5,000.
2. Rebuilt band and segment keys in Power Query and flagged blank balances and impossible ages.
3. Designed a star schema with 8 dimensions and 42 explicit DAX measures in display folders.
4. Tested five management beliefs against the data and wrote each story page around one finding.
5. Built three executive dashboards with synced slicers and a customer drill-through.

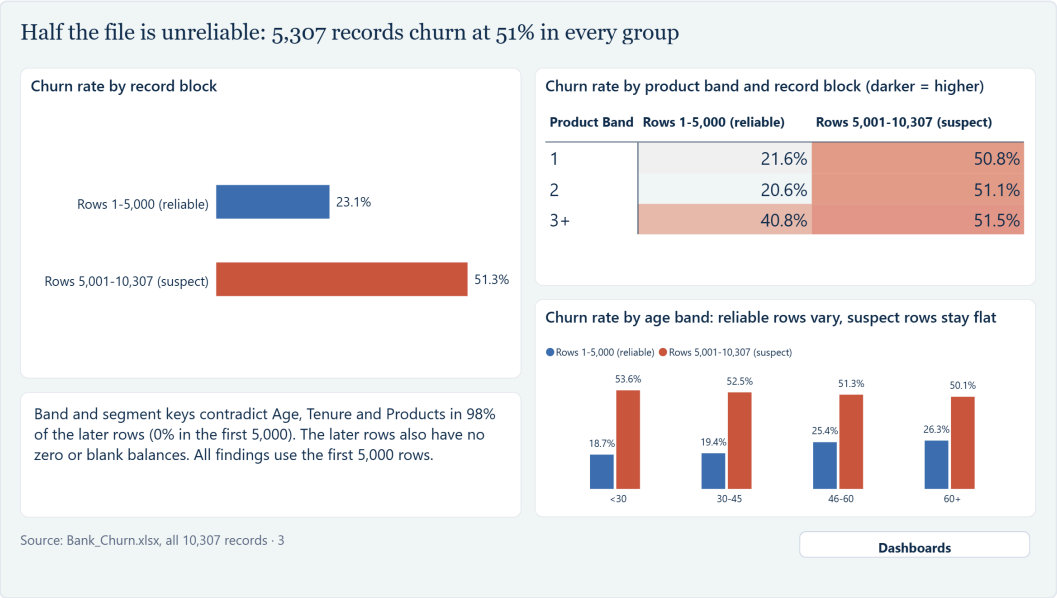
Headline numbers

23.1%
churn rate (1,155 of 5,000)

90.1M
balances held by churners

62%
of churners were inactive

~170
leavers two actions could keep



The scorecard page: headline KPIs and verdicts on each finding.

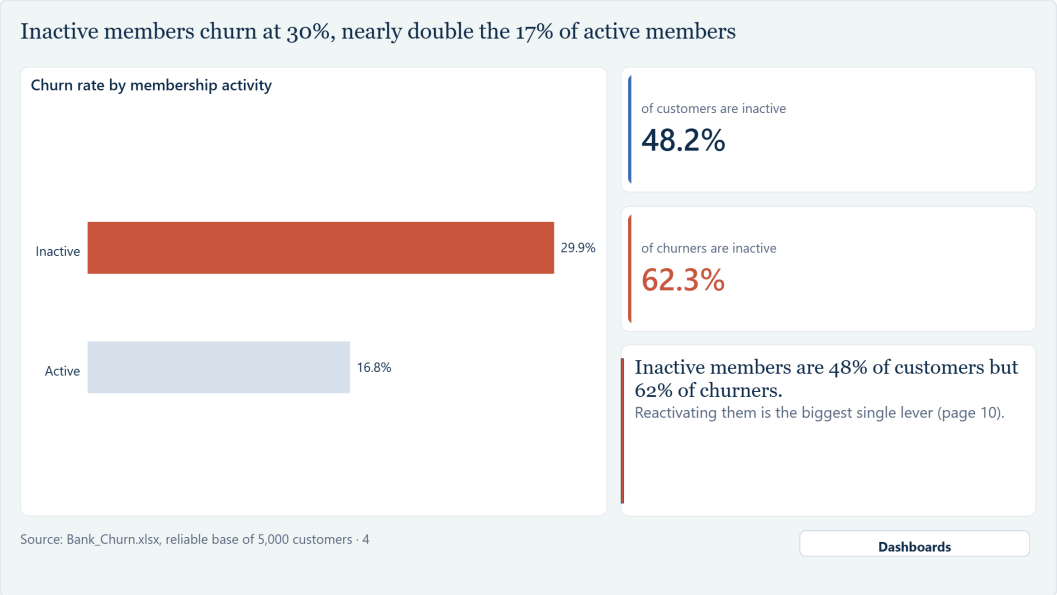
RESULTS

Key insights

Inactivity is the biggest driver	Inactive members churn at 29.9% against 16.8% for active ones.
A third product adds risk	Customers with 3+ products churn at 41%, twice the rate of those with 1-2.
High balance is not safe	High-balance inactive customers churn at 41.8%; high-balance active at 14.6%.
Demographics do not separate leavers	Every country and both genders churn at about 23%.
Half the file is unreliable	5,307 records churn at 51% in every group and were excluded from findings.

Recommendations

1. Reactivate the 2,410 inactive members, starting with the 624 high-balance inactive.
2. Review bundles: stop pushing a third product.
3. Fix the source of the 5,307 suspect records.

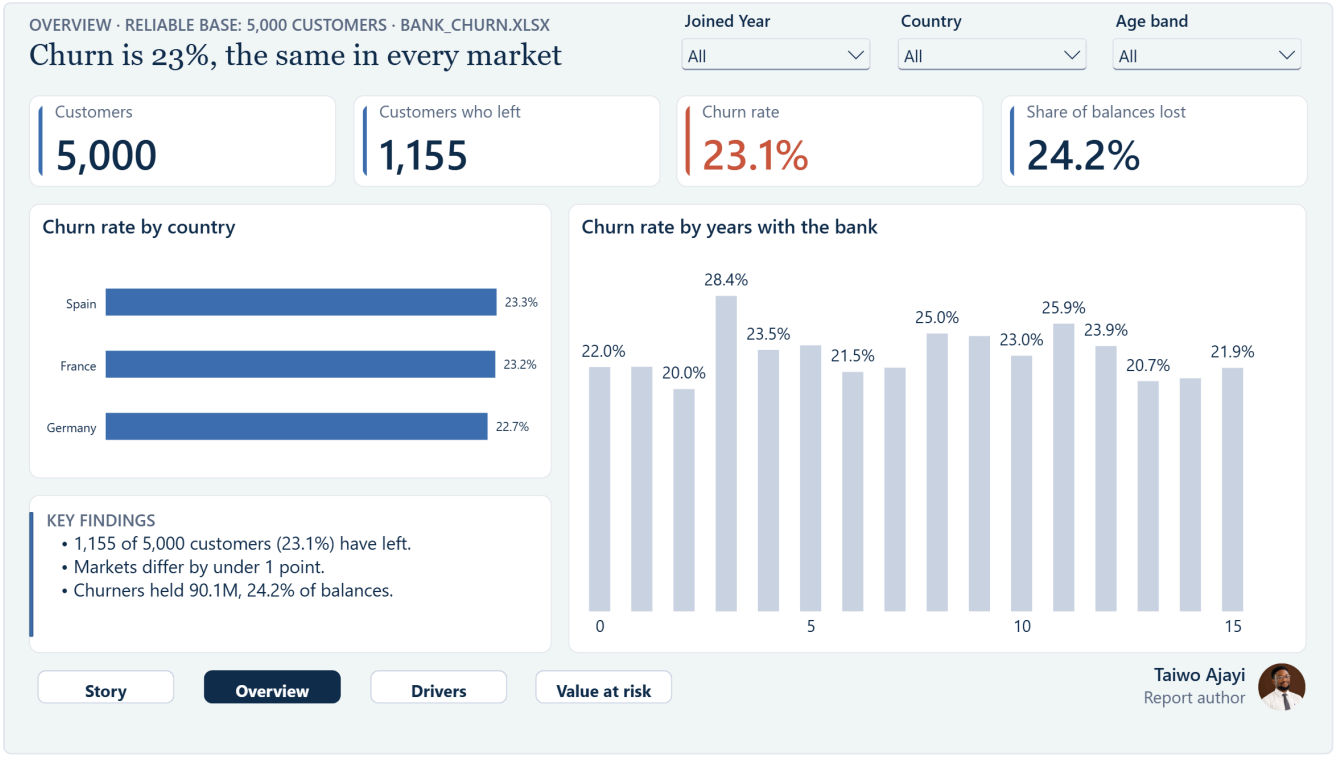


A story page: the title states the finding and the chart shows the evidence.

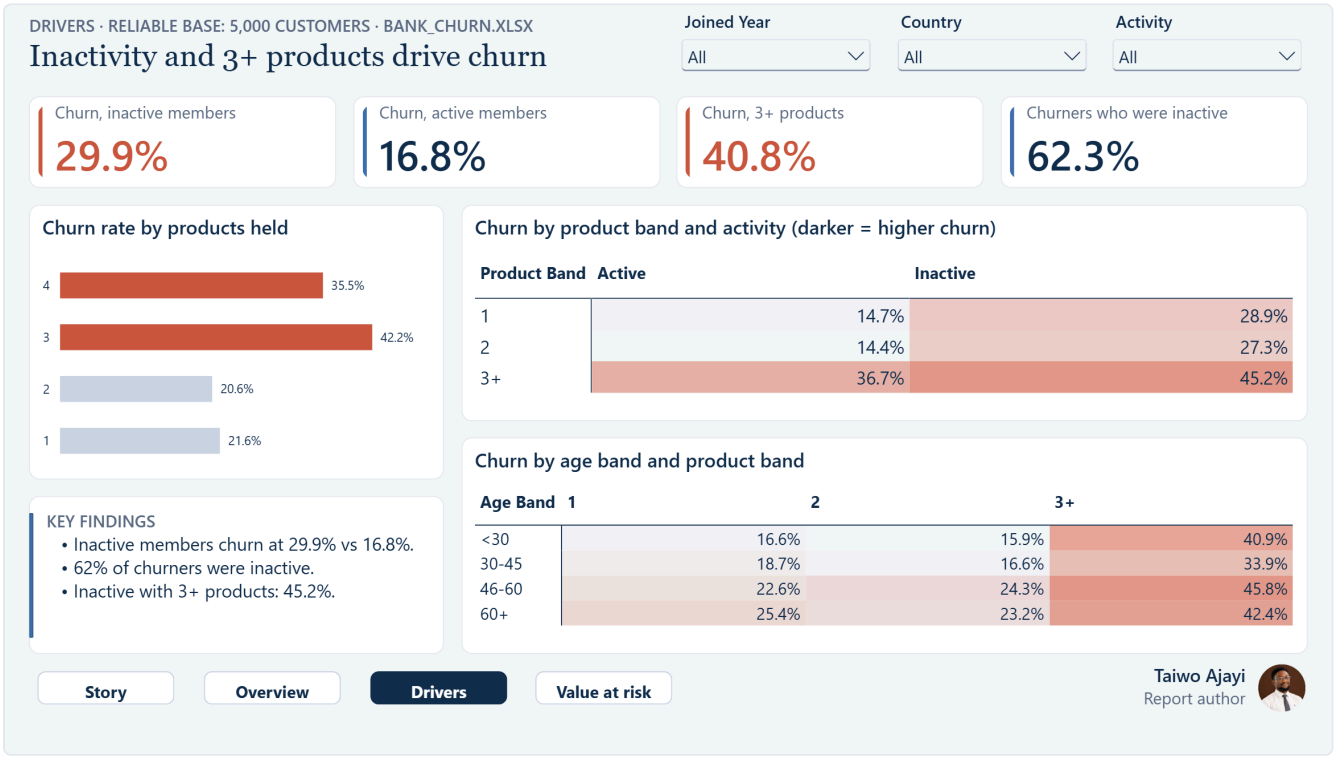
DASHBOARDS

Three interactive dashboards

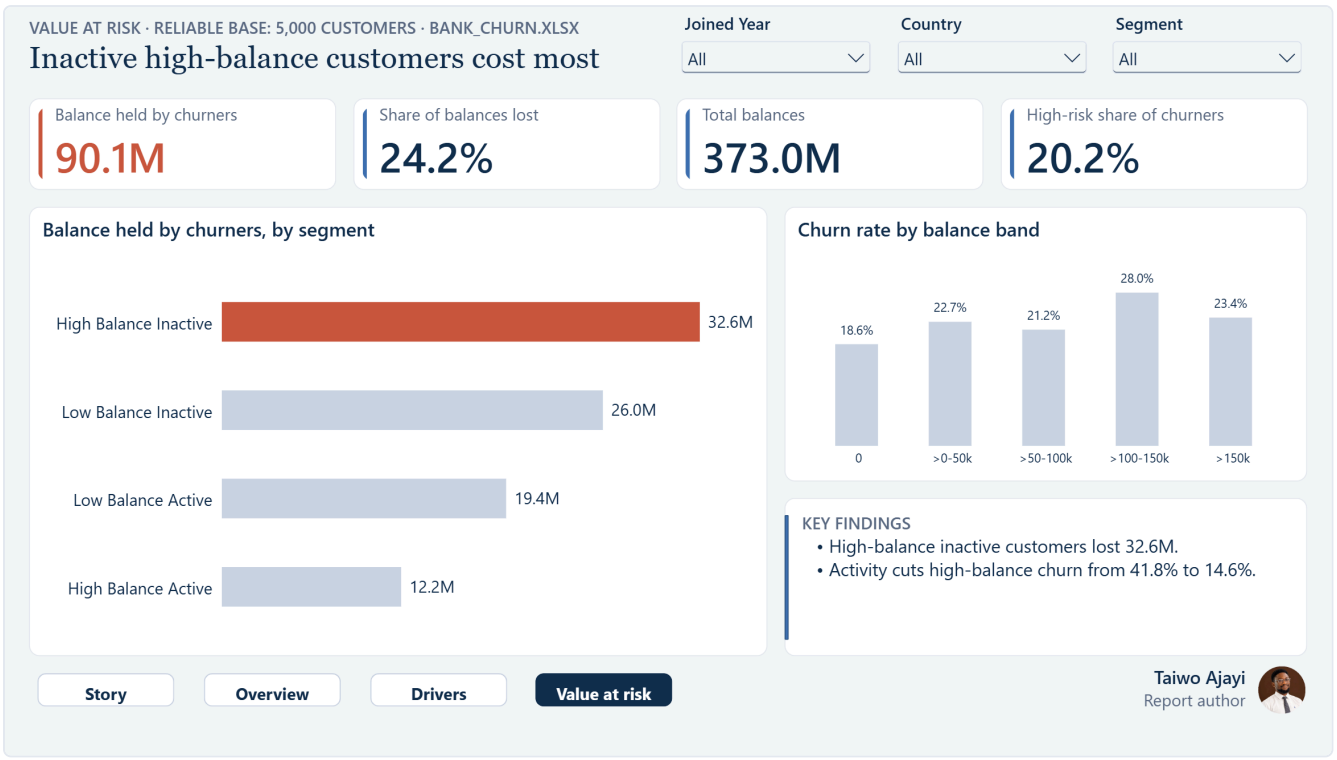
Each dashboard has synced filters, a tab bar to move between them, and a drill-through to a detail page.



D1 Churn overview. Churn by market, tenure and headline KPIs



D2 Customer drivers. Activity and product heat maps



D3 Value at risk. Balances lost by segment



Customer ID	Country	Age	Tenure (years)	Products	Activity	Balance	Churned
4832	Germany	21	4	2	Inactive	238,544	True
2103	Germany	74	12	1	Inactive	220,594	True
3007	France	85	0	2	Inactive	220,161	False
642	France	79	0	2	Active	210,231	False
4608	Spain	47	6	2	Active	209,556	False
2581	France	37	2	1	Inactive	203,607	False
794	Germany	64	14	2	Active	203,134	False
3016	France	49	8	2	Inactive	202,465	True
1521	France	79	3	1	Active	196,544	False
3355	Germany	80	14	1	Active	194,274	False
1136	Germany	81	0	2	Inactive	191,870	True
1464	France	54	9	2	Inactive	191,615	False
1306	France	35	10	1	Inactive	190,688	False
4785	Germany	68	11	1	Active	188,374	False
2808	Germany	75	6	1	Active	184,272	False
4283	Spain	73	6	2	Inactive	182,608	False
2248	Spain	38	5	1	Active	182,024	False

Customer list. Hidden drill-through page: right-click a bar on a dashboard to open it

QUALITY AND SKILLS

What I fixed, and what this shows

Errors I found and corrected in the original analysis

- Separated reliable and suspect records instead of reporting a blended 37.6% churn rate.
- Rebuilt band and segment keys that contradicted the raw values.
- Treated 100 blank balances as zero and flagged them.

Skills demonstrated

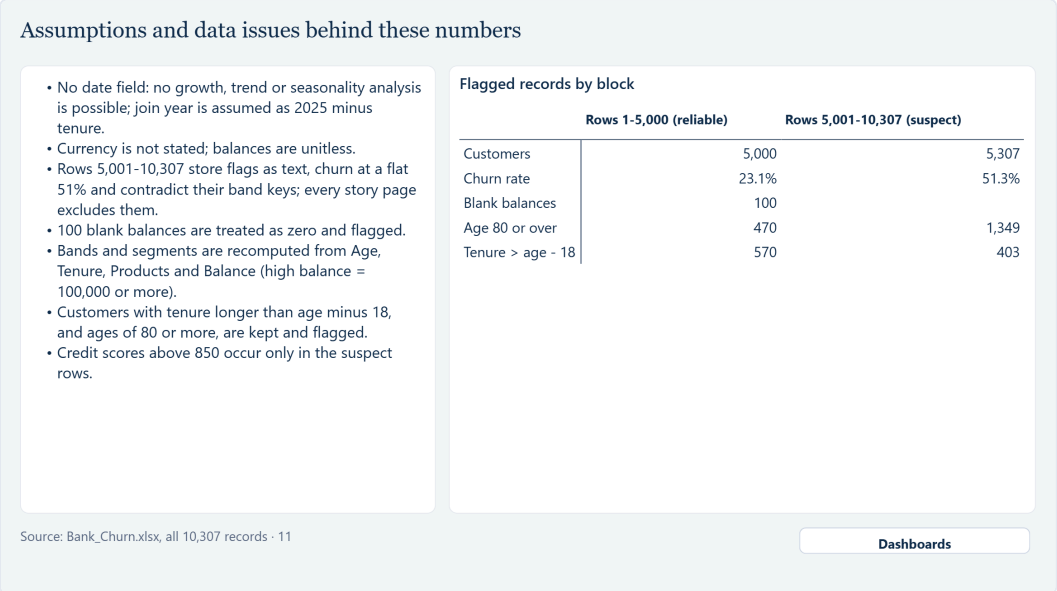
- Data profiling and validation before reporting
- Star-schema modelling and DAX measure design
- Correcting flawed measures and documenting why
- Insight-led storytelling: one finding per page
- Executive dashboard design with drill-through
- Clear documentation for readers and maintainers

Tools

- Power BI Desktop (PBIP project format)
- DAX and Power Query (M)
- TMDL and PBIR source files
- Excel source data

Deliverables

- 15-page Power BI report
- Full report PDF (every page)
- Detailed report guide



The data notes page: every correction and data limit is stated in the report itself.